BofA SECURITIES

Global Memory Tech

# Weekly theme: Meta/CXMT/Korea risks, Japan trip takeaways, Samsung 2Q OP

**Industry Overview**

## Meta's memory chip order cut

Memory chipmakers believe Meta will keep using more actively HBM, LPDDR5, eSSD, etc. for its AI data centers. Thus, the Street's speculation on "Meta will rent overly invested AI servers or cloud infrastructure" should be groundless. In fact, some AI supply chain companies (e.g., NAND controller IC providers, substrate material manufacturers, etc.) said Meta's long-term chip/component orders are getting stronger. Chipmakers said Meta can share its own data centers with external customers as a cloud service provider but this does not translate into excess capacity; it can be a business objective diversification using more advanced memory chips.

## CXMT's DRAM for iPhone

We do not expect Apple to meaningfully adopt CXMT's DRAM. Three reasons: (1) US's restriction on China semis, (2) DRAM quality to meet Apple's specs (10Gbps+ speed, 1.1V power consumption, ECC function), and (3) DRAM IPs owned by major Korean/US chipmakers (litigations if Apple uses less patented chips). True, CXMT can target low-end iPhone 18e but actual DRAM order won't be significant, given weak sales volume of low-end iPhones in China (high-end more popular). Apple may try to use CXMT's DRAM to increase bargaining power when it negotiates 2H or 2027 contract prices with Korea/US big-3 DRAM makers, though actual purchasing volume (from CXMT) remains small.

## Peak concern following Korea's cluster construction plan

Some investors said the Korea government-led new memory fab cluster construction spending W800tn in the country's southwestern region (e.g., Kwangju) signals a peak of memory cycle. However, we do not expect meaningful chip production volumes by 2033 (see report). It is a long-term plan after focusing on Youngin/Pyeongtaek fab cluster expansion (2026-35). Overall, we don't see yet any capex-driven memory cycle peak, given strong/diversified enterprise chip demand (HBM, SOCAMM, eSSD, etc.).

## What we learned from our Japan trip

We met Japan investors and NAND supply chain companies this week. The investors were broadly bullish on memory but frequently asked potential downturn given currently strong industry growth. In fact, corporate management provided more bullish guidance, suggesting (1) upbeat 2Q ASP (particularly NAND), (2) still rising trend of 3Q/4Q ASP vs 2Q, (3) 2027 shortage (DRAM, NAND), (4) rising LTA (but still volume-centric), (5) disciplined capex and chip production volume (mainly for Japan NAND).

## Samsung's 2Q preliminary OP

Samsung Electronics' 2Q preliminary OP (due 7 July) is likely to be slightly lower than the bullish consensus due to special bonus (including 1Q) and smartphone margin squeeze. Memory-only OP can exceed consensus due to upbeat ASP – consistent with Korea's strong exports. DRAM spot/contract price also rebounded further in June.

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**Exhibit 1: DDR5 stable, DDR4 rebounded for four consecutive weeks; NAND flat or slightly down**

Spot prices – DRAM and NAND

| US$ | Current | WoW | QoQ | YoY |
|---|---|---|---|---|
| **DRAM spot** | | | | |
| 16Gb DDR5 | 46.8 | 0% | 27% | 671% |
| 16Gb DDR4 | 75.4 | 5% | 3% | 783% |
| 8Gb DDR4 | 36.3 | 1% | 8% | 616% |
| 4Gb DDR4 | 11.9 | 6% | 69% | 384% |
| **NAND spot** | | | | |
| 1Tb wafer | 24.4 | 0% | -10% | 381% |
| 512Gb wafer | 19.9 | -1% | -10% | 641% |
| 256Gb wafer | 10.2 | 0% | -7% | 578% |

**Source:** DRAMeXchange

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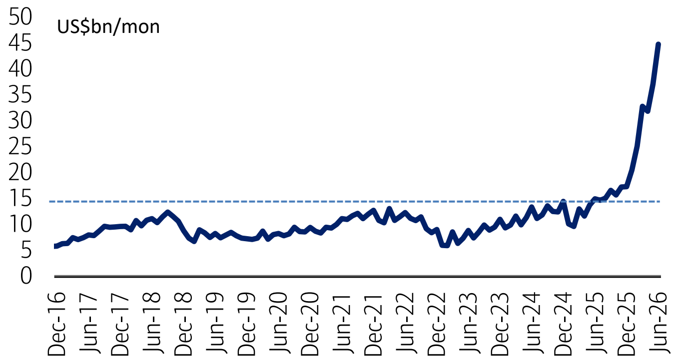
**AI**: Artificial intelligence
**ASP**: Average selling price
**CXMT**: ChangXin Memory Technologies
**DDR4/5**: 4th/5th gen double-data rate DRAM
**DRAM**: Dynamic random-access memory
**ECC**: Error correction code
**Gbps:** Gigabit per second
**HBM**: High bandwidth memory
**IC**: Integrated circuit
**IP:** Intellectual property
**LPDDR5**: Low-power DDR5
**LTA**: Long-term agreement
**NAND**: Not-AND memory
**SOCAMM**: Small-outline compression attached memory module

# Korea exports and TrendForce memory ASP forecast

**Exhibit 2: Significant jump in Jun (US$44.8bn; +21% MoM); more than 3x higher than 2025-average (US$14bn)**
Korea semis exports - US$bn/month




**Source:** MoTIR

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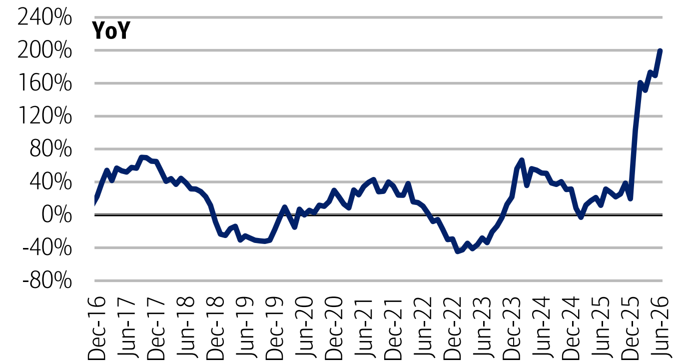
**Exhibit 3: YoY rebound surged to +199% in Jun; already six consecutive months of triple-digit growth**
Korea semis exports - YoY change in monthly US$bn




**Source:** MoTIR

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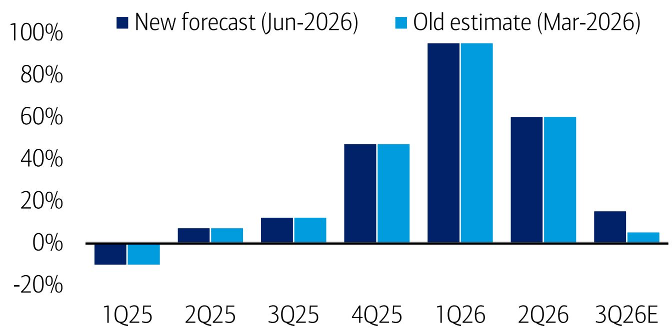
**Exhibit 4: TrendForce has revised up its 3Q26 DRAM ASP forecast to 13–18% QoQ vs previous estimate of 3–8% QoQ**
TrendForce DRAM ASP forecast (QoQ change) – Jun-26 update vs Mar-26




**Source:** TrendForce

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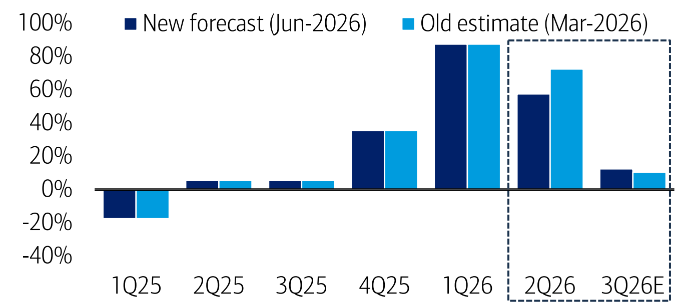
**Exhibit 5: TrendForce has revised down its 2Q26 NAND ASP forecast to 55–60% QoQ (from 70–75%), while raising its 4Q26 outlook to 10–15% QoQ from the earlier 8–13% range**
TrendForce NAND ASP forecast (QoQ change) – Jun-26 update vs Mar-26




**Source:** TrendForce

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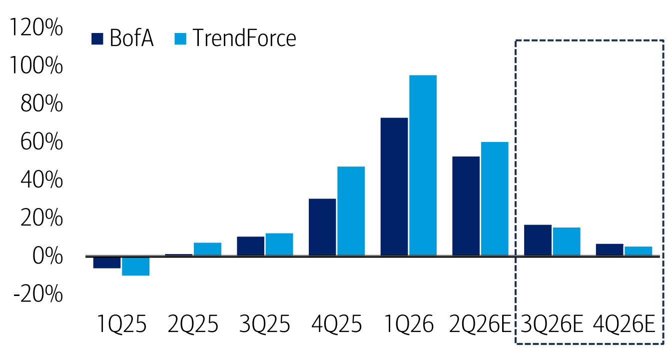
**Exhibit 6: Our estimates are largely in line for 3Q26 and 4Q26 and more conservative for 2Q26 relative to TrendForce, with DRAM ASPs projected to increase 53% / 17% / 7% QoQ across 2Q–4Q26 (BofA)**
BofA vs TrendForce DRAM ASP assumptions (QoQ change)




**Source:** TrendForce, BofA Global Research estimates

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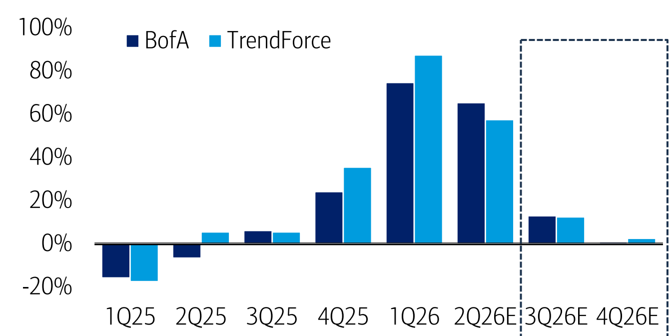
**Exhibit 7: Similarly, our NAND assumptions are either conservative or broadly aligned with TrendForce, with ASPs expected to grow 65% / 13% / 1% QoQ across 2Q–4Q26**
BofA vs TrendForce NAND ASP assumptions (QoQ change)




**Source:** TrendForce, BofA Global Research estimates

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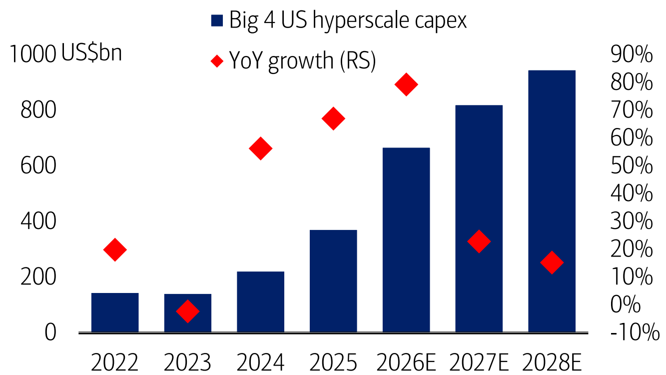
# Hyperscale capex and cloud revenue/margins

**Exhibit 8: Amazon, Microsoft, Alphabet, and Meta keep increasing capex spend - ~80% YoY in 2026 to reach ~US$0.7tn; near-US$1tn annual capex expected in 2027-28E; Meta's growth also on track**
Combined capex of top 4 US tech companies




**Source:** Company reports, BofA Global Research estimates

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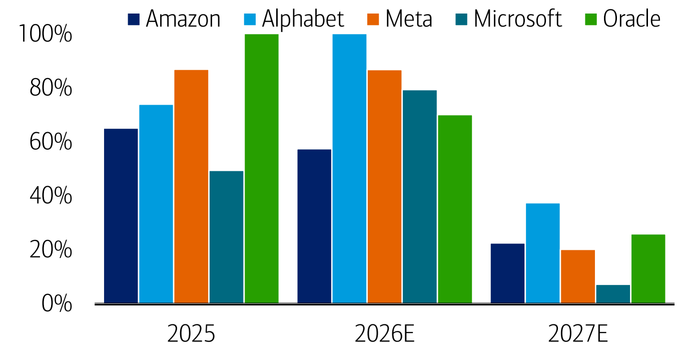
**Exhibit 9: Record-high capex spend by all US hyperscalers expected to continue in 2026-27E; no capex cut in 2027**
Hyperscalers' capex growth – YoY change




Note: Microsoft and Oracle data adjusted for CY basis **Oracle's 2025 capex up 230% YoY MSFT capex for 2026-27 is based on consensus
**Source:** Companies, BofA Global Research estimates

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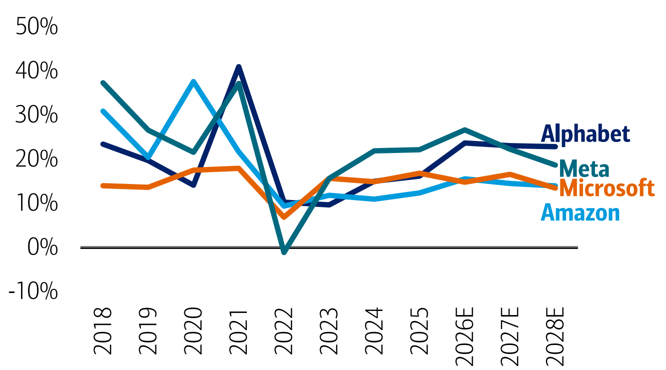
**Exhibit 10: Amazon, Microsoft, Alphabet, and Meta overall revenue expected to grow +15-20% in 2026-28**
Overall revenue growth trend of top 4 US hyperscalers




**Source:** Company reports, BofA Global Research estimates

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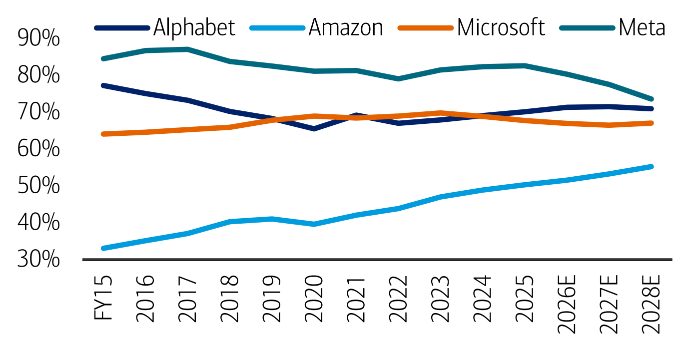
**Exhibit 11: GM of hyperscalers strong at around 70-80% levels**
Gross margin trend of top 4 US hyperscalers




**Source:** Company reports, BofA Global Research estimates

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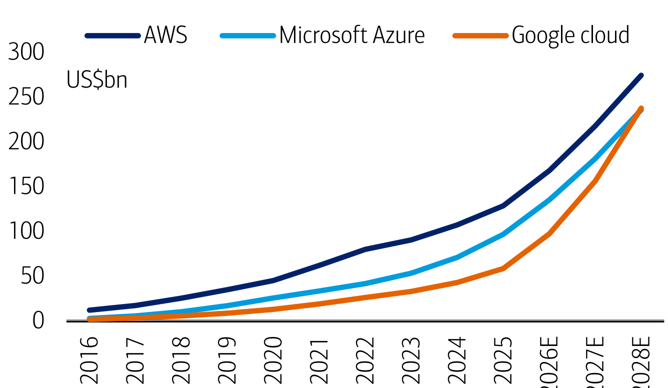
**Exhibit 12: Cloud revenue growth still solid at 35-40% YoY in 2026-28**
Cloud revenue of hyperscalers consistently strong




**Source:** Company reports, BofA Global Research estimates

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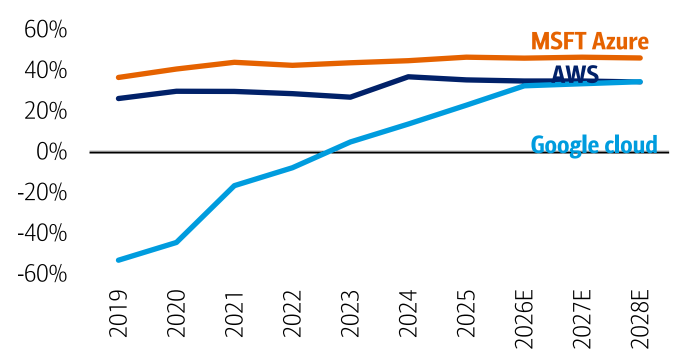
**Exhibit 13: High OPM seen among AWS (35%+) and Azure (40%+); Google's margin also set to reach 30-35% in 2026-28**
Cloud OPM trend also high




**Source:** Company reports, BofA Global Research estimates

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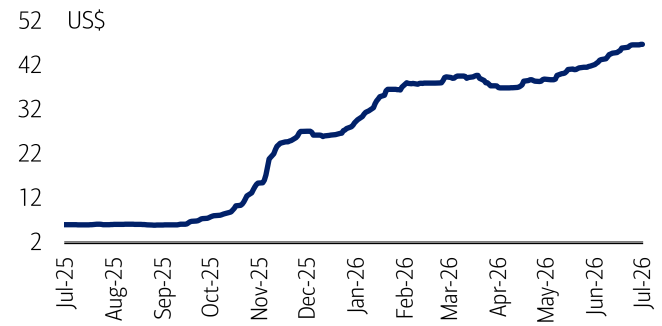
# Memory spot price short term trend

**Exhibit 14: DDR5 prices reached new highs ($47) in early-July, supported by strong AI-related demand and constrained supply as DRAM makers prioritize HBM production**
16Gb DDR5 spot price – daily, Jul'25 - Jul '26




**Source:** DRAMeXchange

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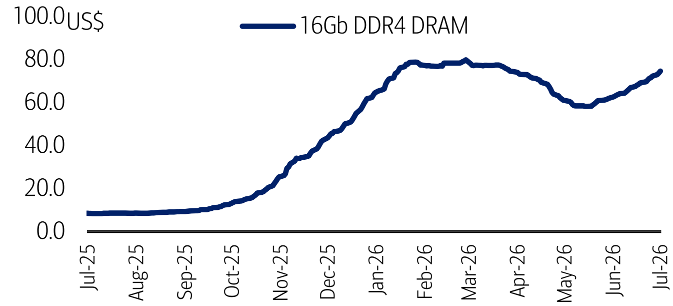
**Exhibit 15: DDR4 prices remained near cycle highs, supported by supply cuts and customer restocking, with a brief correction in May before recovering through June and early-July**
16Gb DDR4 spot price trend, Jul'25 - Jul '26




**Source:** DRAMeXchange

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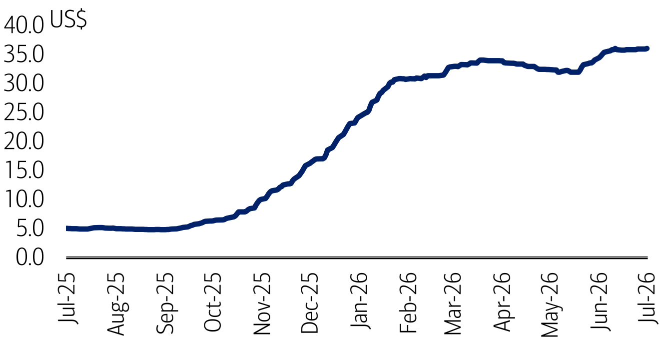
**Exhibit 16: 8Gb DDR4 prices remained firm near highs, driven by production exits and inventory replenishment**
8Gb DDR4 spot price trend, Jul'25 - Jul '26




**Source:** DRAMeXchange

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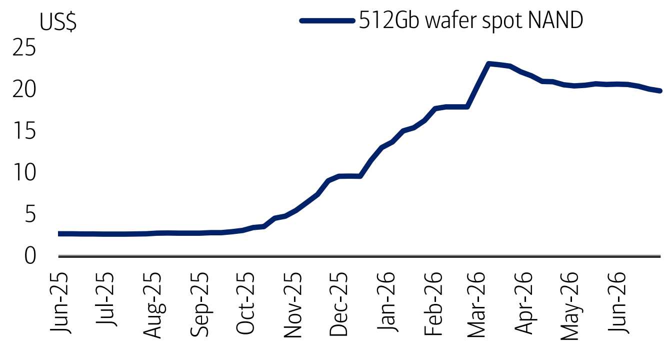
**Exhibit 17: NAND wafer pricing rose to elevated levels, through 4Q25-1Q26, on tight supply and stronger SSD demand, but has recently flattened/slightly softened**
512Gb NAND wafer spot price – weekly, Jun'25 - Jun '26




**Source:** DRAMeXchange

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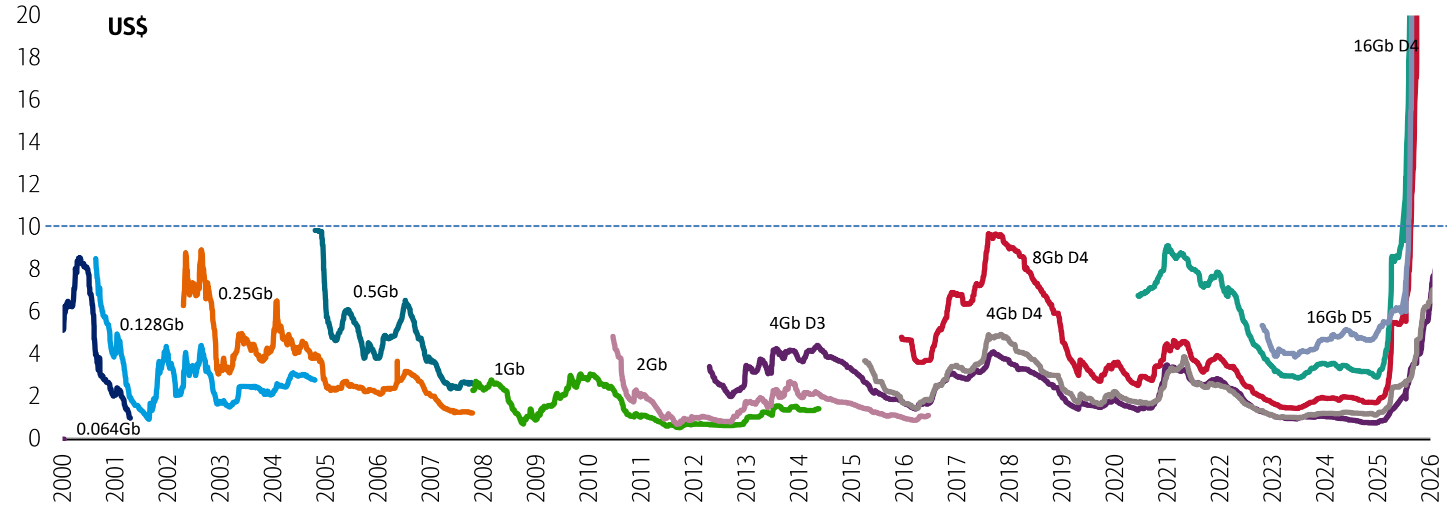
## Memory spot/contract price long term trend

**Exhibit 18: DRAM spot prices rebounded through June after softening through April–May, following a strong rally from September 2025 to January 2026. Prices have reached multi-decade highs, with 16Gb DDR5 at ~$47 and DDR4 at ~$75, driven by supply reallocation toward HBM and server DRAM—well above the prior ~$10 peak in October 2017.**

DRAM spot price – long-term trend (2000-2025)




**Source:** DRAMeXchange

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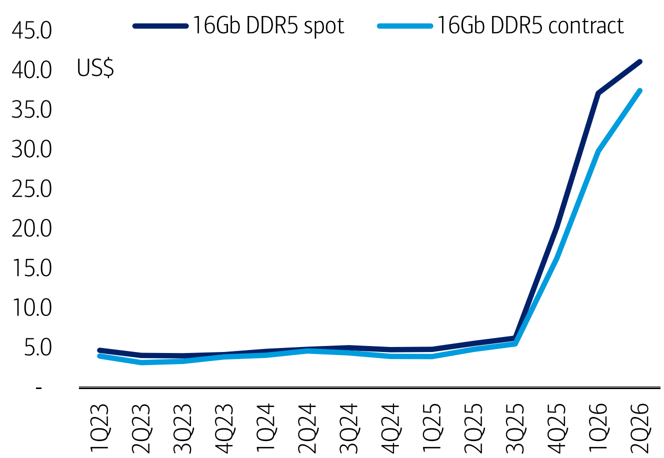
**Exhibit 19: DRAM spot and contract prices climbed to record highs of ~$35–40, with moderate growth in 2Q26 following a strong rally in 4Q25 and 1Q26.**

DRAM spot and contract price - quarterly average trend




**Source:** DRAMeXchange, TrendForce

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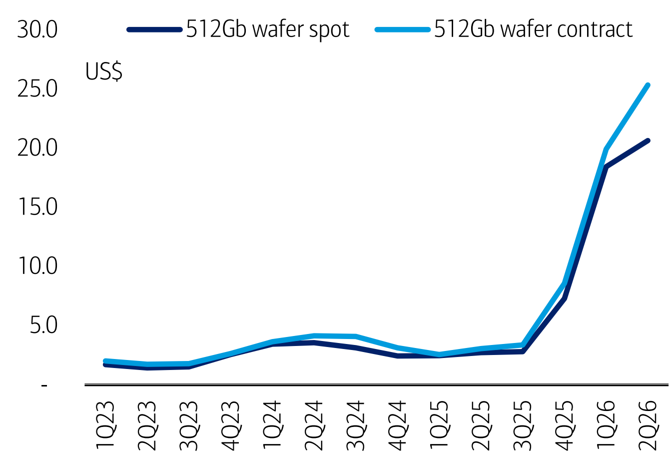
**Exhibit 20: NAND wafer contract prices have surged significantly ahead of spot prices, with both now at record-high levels**

NAND wafer spot and contract price - quarterly average trend




**Source:** DRAMeXchange, TrendForce

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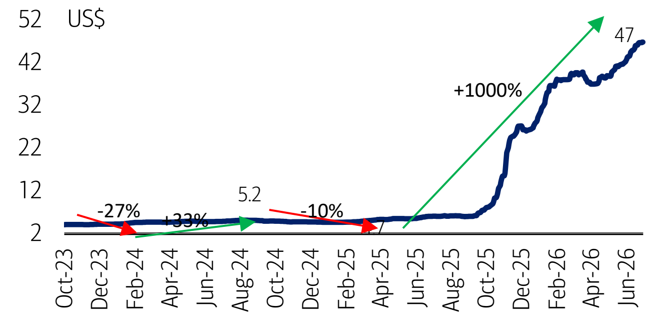
**Exhibit 21: Prices rebounded through May and June, reaching an all-time high ($47) following a volatile April. Uptrend was driven by strong gains in Oct (+70%), Nov (+60%), Jan (+25%), before moderating into a more subdued trend during Feb–Mar**

16Gb DDR5 spot price – daily, Oct '23 - Jun '26




**Source:** DRAMeXchange

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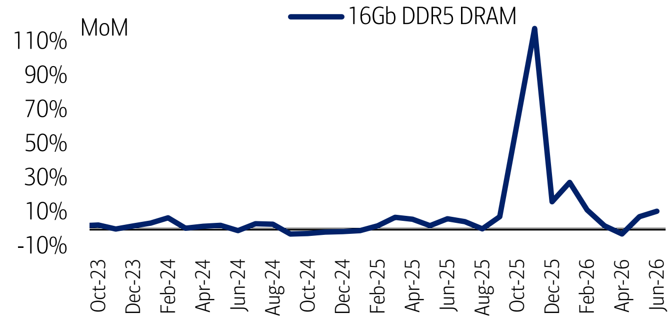
**Exhibit 22: Modest uptick observed in May/Jun (+7-10%), following a sequential slowdown from Feb (+10%) to flat momentum in Mar and a minor decline in Apr, after the strong Jan (+25%) surge and the sharp Oct–Nov-25 rally (+60–120%)**

16Gb DDR5 spot month-average price – MoM change, Oct '23 - Jun '26




**Source:** DRAMeXchange

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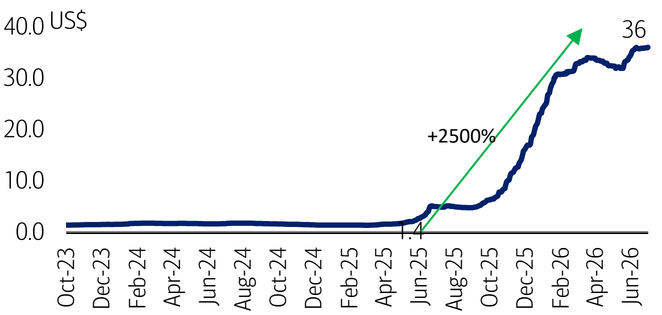
**Exhibit 23: 8Gb DDR4 prices remained flat in 2H-Jun vs up in 1H-Jun and through May after stabilizing in April; they remain significantly higher YoY due to supply cuts by major vendors, currently around $30—well above the prior ~$10 peak in October 2017**

8Gb DDR4 spot price trend, Oct '23 - Jun '26




**Source:** DRAMeXchange

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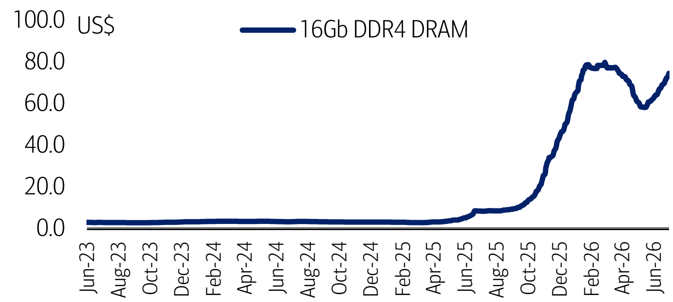
**Exhibit 24: Prices edged up slightly in late May/through June after a ~25% correction from April to mid-May, having peaked near ~$80 in early March. Despite this pullback, prices remain elevated—up ~2000% from ~$3 in October 2025—supported by strong gains through Sep–Dec (+30–70%) and January (+20%)**

16Gb DDR4 spot price trend, Jun '23 - Jun '26




**Source:** DRAMeXchange

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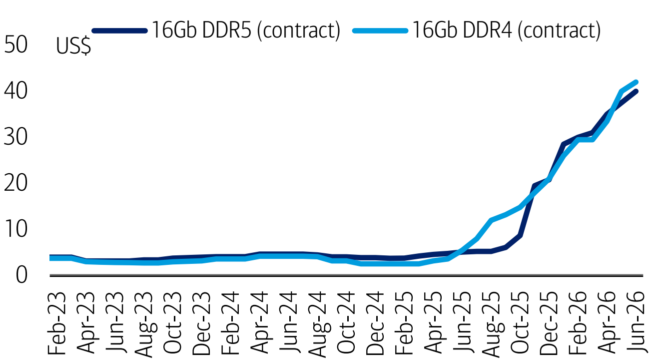
**Exhibit 25: DDR4 and DDR5 (16Gb) contract prices are similar at US$35-$40 levels – DDR5 price premium no longer exists due to DDR4 shortage**

16Gb DDR5 vs 16Gb DDR4 contract price trend, Feb '23-Jun '26




**Source:** TrendForce

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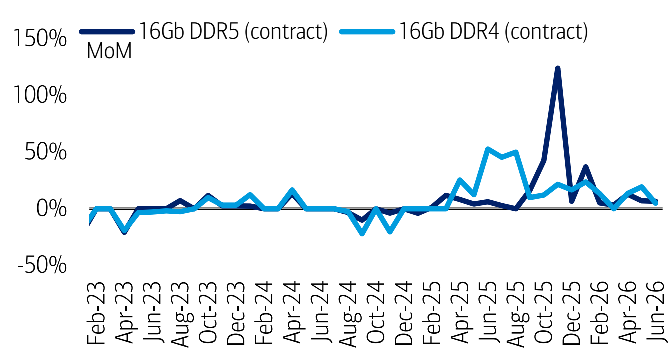
**Exhibit 26: DDR5 and DDR4 contract prices expected to be flat MoM in Jun vs up +10-15% MoM each in Apr/May-26; price strong through 2H25 and early 2026**

16Gb DDR5 vs 16Gb DDR4 contract price change - MoM, Feb '23-Jun '26




**Source:** TrendForce

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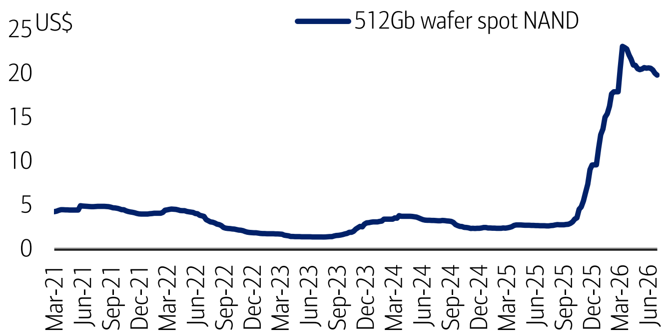
**Exhibit 27: NAND spot prices broadly stable in 2H-May and softened through-June, after trending down gradually through late March, April, and 1H-May, still up 50%+ YTD and 8x vs Feb-25 low ($2.4)**
512Gb NAND wafer spot price – weekly, Mar '21 - Jun '26




**Source:** DRAMeXchange

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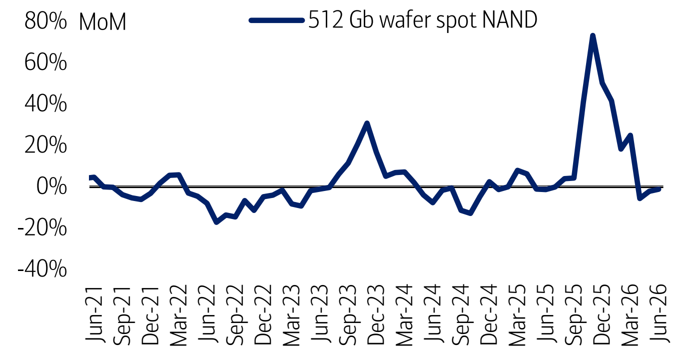
**Exhibit 28: Flat in June, while slightly corrected in Apr/May vs up +15-20% in Feb/ Mar-26 and up +40-70% MoM in Oct/Nov/Dec/Jan**
512Gb NAND wafer spot average – MoM change, Jun'21 - Jun'26




**Source:** DRAMeXchange

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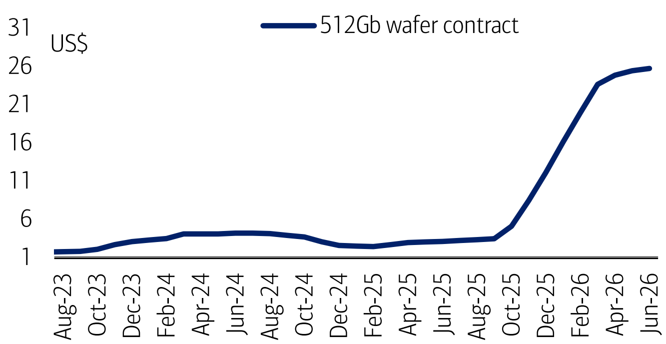
**Exhibit 29: Current contract price at ~$25, around 10x vs Feb-25 bottom of $2.5; prices significantly up in 4Q25/1Q26 and remained broadly stable thereafter**
NAND wafer contract price trend, Aug '23-Jun '26




**Source:** TrendForce, DRAMeXchange, BofA Global Research

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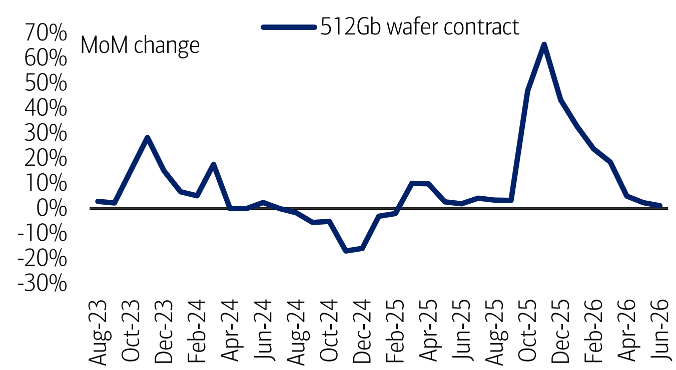
**Exhibit 30: Apr/May/Jun price only up +1-5% MoM each following Jan/Feb/Mar-26 rally (up 20-30% MoM) and Oct/Nov/Dec upturn (up 40-60% MoM)**
MoM change of NAND contract price (512Gb wafer Aug '23-Jun '26




**Source:** DRAMeXchange, TrendForce, BofA Global Research

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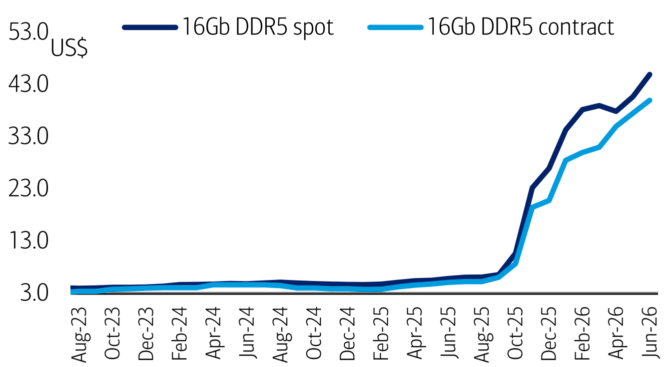
**Exhibit 31: 16Gb DDR5 spot and contract prices are in the range of US$35-40 vs historical range of US$3-5**
16Gb DDR5 spot vs contract price trend (US$), Aug '23-Jun '26




**Source:** DRAMeXchange, TrendForce

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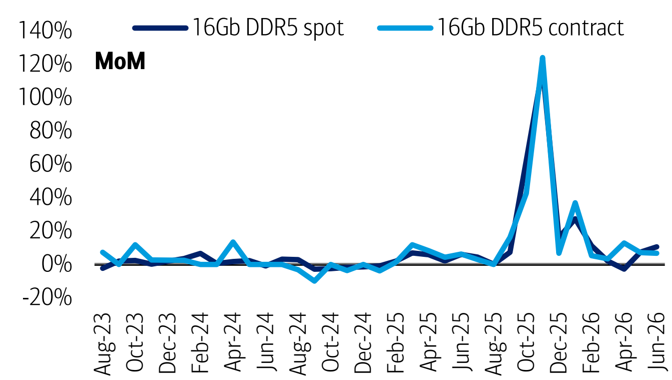
**Exhibit 32: Jun'26 spot and contract prices turned positive again after Feb-Mar moderation, prices had witnessed a sharp Oct–Dec'25 rally**
16Gb DDR5 spot vs contract price trend – MoM change, Aug '23-Jun '26




**Source:** DRAMeXchange, TrendForce

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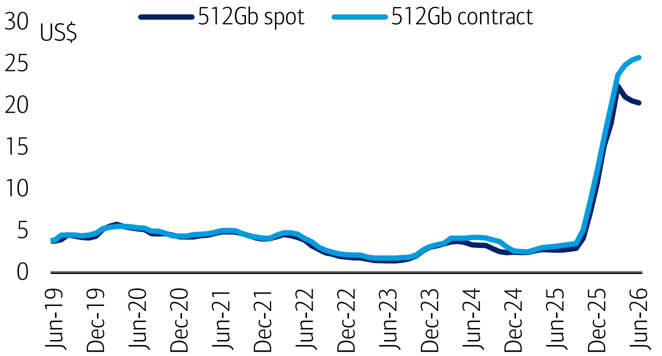
**Exhibit 33: Current NAND spot and contract prices are a few times higher than 2025 summer level**
512Gb NAND wafer spot vs contract price trend (US$), Jun '19 - Jun '26




**Source:** DRAMeXchange, TrendForce

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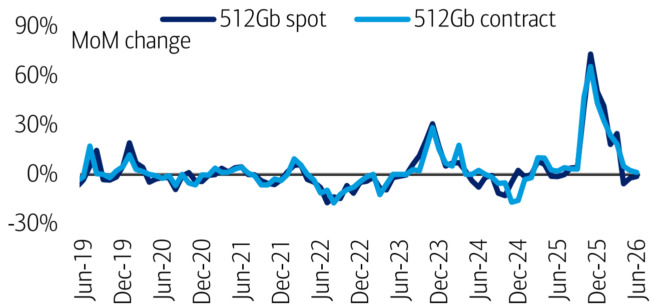
**Exhibit 34: NAND spot and contract prices normalized in Apr/May/Jun after strong rally during Oct-25 – Dec-25 (up 40-50% per month) and Jan-Mar-26 (20-30% per month)**
512Gb NAND spot vs contract price trend – MoM change, Jun '19 - Jun '26




**Source:** DRAMeXchange, TrendForce

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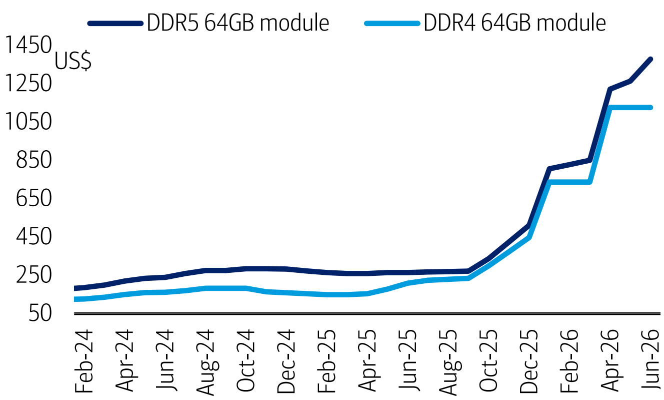
**Exhibit 35: Current price of 64GB DDR4/DDR5 modules hit all-time high of more than $1,000 level (DDR5: US$1,400, DDR4: US$1,100)**
Server DRAM contract price trend – DDR5 vs DDR4 modules, Feb '24-Jun '26




**Source:** TrendForce

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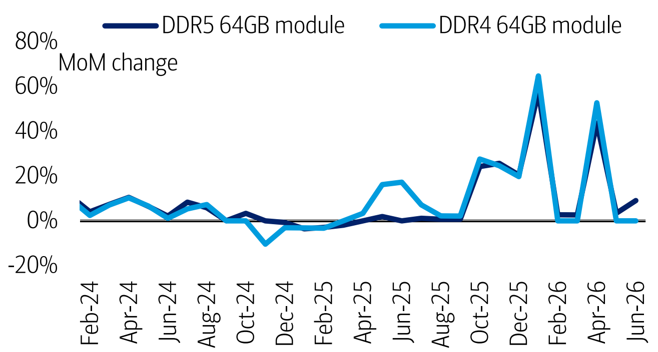
**Exhibit 36: DDR5 server contract prices recovered in June, while DDR4 remained flat in, following a sharp 40–50% MoM increase in Apr-26 and a similar 50–60% surge in January, after largely stable pricing conditions during Feb–Mar**
MoM change of server DRAM contract prices, Feb '24-Jun '26




**Source:** TrendForce

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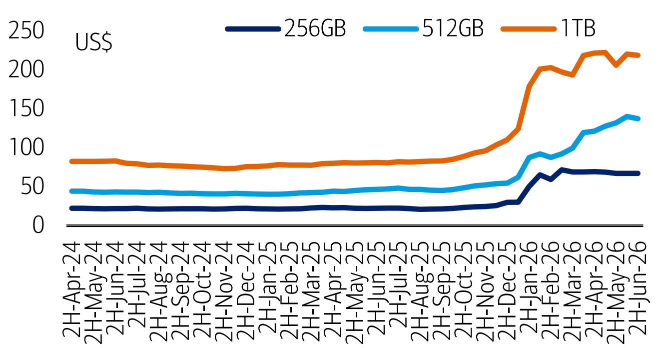
**Exhibit 37: 2H-Jun SSD price slightly down vs 1H-Jun rebound; prices witnessed a sharp surge in 1Q26 and April (DRAMeXchange), following a gradual upward trend throughout 2025**
Client SSD price trend – mostly for PC (not server), Apr '24 - Jun '26




Note: SSD prices are as of 28 Jun 2026, reported by DRAMeXchange. TB = terabyte.
**Source:** DRAMeXchange

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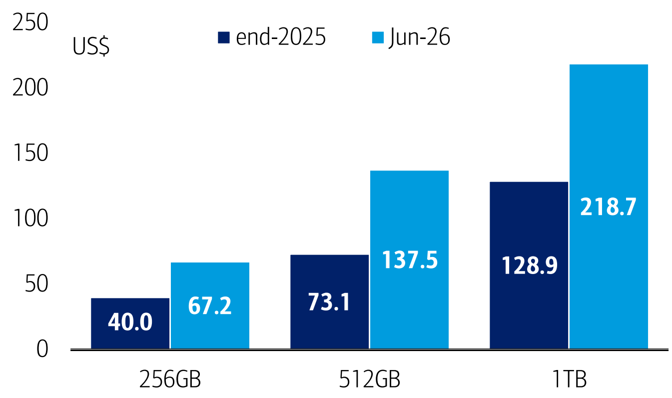
**Exhibit 38: Jun prices have doubled compared to end-2025 levels, while the 2025 increase was more modest at around 35–40%**
Client SSD (for PC) price comparison – current vs. end-2025




Note: SSD prices are as of 28 Jun 2026, reported by DRAMeXchange
**Source:** DRAMeXchange

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## Valuation and stock performance

**Exhibit 39: Memory stocks sharply corrected on 2 July; we believe Meta/CXMT/Korea were newly overly concerned vs solid fundamentals (still rising AI capex spend, CXMT's DRAM for Apple likely not meaningful, low impact of Korea's new memory cluster by 2030, etc); P/E multiples still low despite robust stock price rally in May/June**

Valuation comparison among memory and semiconductor supply chain stocks

| | Ticker | Rating | Price | Mcap | P/E | | | P/Book | | | EV/EBITDA | | EV/Sales | | ROE | | | Div. yield | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | (Local) | ($bn) | FY25 | FY26E | FY27E | FY25 | FY26E | FY27E | FY26E | FY27E | FY26E | FY27E | FY25 | FY26E | FY27E | FY26E | FY27E |
| **Memory/DRAM** | | | | | | | | | | | | | | | | | | | |
| SK Hynix | HXSCF | RSTR** | 2,551,000 | 1,025.0 | - | - | - | 15.1 | - | - | - | - | - | - | - | - | - | - | - |
| Samsung | SSNLF | B-1-7 | 316,000 | 1,082.4 | 48.1 | 8.2 | 6.6 | 5.1 | 3.2 | 2.2 | 4.9 | 4.0 | 2.6 | 2.1 | 10.8% | 47.0% | 39.1% | 0.9% | 1.5% |
| Micron | MU | C-1-7 | 1,032.28 | 1,165.8 | 124.5 | 14.1 | 7.4 | 21.5 | 8.6 | 4.4 | 11.4 | 6.0 | 8.7 | 4.9 | 18.8% | 87.8% | 81.0% | 0.1% | 0.1% |
| Nanya | NNYAF | B-1-7 | 407.00 | 44.1 | 192.1 | 6.5 | 4.8 | 8.2 | 3.9 | 2.5 | 4.7 | 3.4 | 3.5 | 2.5 | 3.9% | 73.9% | 57.6% | 4.9% | 6.1% |
| **Memory/NAND** | | | | | | | | | | | | | | | | | | | |
| Kioxia | XIYUF | C-1-9 | 76,260 | 257.0 | 74.5 | 8.3 | 6.4 | 29.5 | 6.5 | 3.2 | 5.2 | 4.0 | 4.2 | 3.3 | 51.9% | 128.2% | 67.2% | 0.0% | 0.0% |
| Sandisk | SNDK | C-1-9 | 2,032.22 | 301.0 | n/a | 29.5 | 8.7 | 32.7 | 16.0 | 5.2 | 22.6 | 6.5 | 14.5 | 5.5 | 4.3% | 76.0% | 96.8% | 0.0% | 0.0% |
| Western Digital | WDC | C-1-7 | 598.37 | 219.0 | n/a | 60.1 | 32.0 | 41.2 | 17.7 | 11.8 | 40.6 | 22.1 | 17.0 | 11.7 | 22.5% | 42.9% | 46.5% | 0.1% | 0.1% |
| SIMO | SIMO | C-1-7 | 316.98 | 10.7 | 87.1 | 35.2 | 29.8 | 12.6 | 9.7 | 7.8 | 26.2 | 20.6 | 6.1 | 5.2 | 15.3% | 31.9% | 29.6% | 0.8% | 0.9% |
| Phison | PISNF | C-1-8 | 2,240.00 | 15.5 | 56.4 | 6.2 | 10.6 | 8.1 | 4.2 | 3.5 | 4.6 | 7.5 | 1.9 | 1.7 | 16.0% | 89.2% | 36.0% | 8.5% | 5.1% |
| **US/Taiwan semis** | | | | | | | | | | | | | | | | | | | |
| NVIDIA | NVDA | C-1-7 | 197.58 | 4,781.4 | 43.4 | 21.7 | 14.9 | 31.3 | 16.2 | 10.2 | 18.1 | 12.6 | 12.3 | 8.6 | 94.3% | 95.9% | 80.3% | 0.4% | 0.6% |
| AMD | AMD | C-1-9 | 540.88 | 882.0 | 129.7 | 72.1 | 40.8 | 14.0 | 11.9 | 9.1 | 52.0 | 31.3 | 17.7 | 11.4 | 11.3% | 18.1% | 26.3% | 0.0% | 0.0% |
| Broadcom | AVGO | C-1-7 | 369.34 | 1,757.2 | 54.2 | 31.8 | 20.6 | 21.7 | 16.7 | 10.7 | 25.3 | 16.1 | 16.8 | 11.0 | 45.3% | 61.3% | 65.5% | 0.7% | 0.8% |
| Intel | INTC | C-1-9 | 127.02 | 638.4 | 295.4 | 119.8 | 73.8 | 5.1 | 4.6 | 4.3 | 31.6 | 24.0 | 10.6 | 9.2 | 1.7% | 4.1% | 6.2% | 0.0% | 0.0% |
| Arm | ARM | C-2-9 | 337.47 | 360.4 | 190.7 | 163.0 | 118.8 | 43.5 | 34.4 | 26.6 | 128.8 | 92.7 | 60.7 | 45.4 | 25.0% | 23.6% | 25.5% | 0.0% | 0.0% |
| Qualcomm | QCOM | B-3-7 | 181.92 | 191.7 | 15.1 | 17.1 | 16.6 | 9.0 | 7.2 | 6.9 | 14.1 | 14.2 | 4.9 | 4.7 | 56.0% | 47.6% | 41.7% | 2.0% | 2.0% |
| TSMC | TSMWF | B-1-7 | 2,465 | 2,005.2 | 37.2 | 24.0 | 16.4 | 11.8 | 8.4 | 5.8 | 15.5 | 10.9 | 11.4 | 8.2 | 35.4% | 40.8% | 41.8% | 1.0% | 1.0% |
| MediaTek | MDTKF | B-1-7 | 4,345 | 218.6 | 65.7 | 66.8 | 31.9 | 17.4 | 18.4 | 12.7 | 52.4 | 25.8 | 10.6 | 6.4 | 26.4% | 26.6% | 46.8% | 1.2% | 2.5% |
| **Equipment** | | | | | | | | | | | | | | | | | | | |
| ASML | ASMLF | B-1-7 | 1,641.60 | 726.0 | 66.2 | 49.5 | 35.1 | 32.5 | 26.5 | 21.7 | 39.2 | 28.5 | 15.7 | 12.5 | 50.6% | 58.7% | 67.5% | 0.5% | 0.6% |
| AMAT | AMAT | B-1-7 | 650.91 | 516.8 | 69.1 | 53.6 | 40.1 | 25.3 | 17.6 | 12.4 | 45.7 | 34.2 | 15.6 | 12.5 | 38.6% | 39.0% | 36.0% | 0.3% | 0.4% |
| Lam | LRCX | C-1-7 | 391.26 | 489.3 | 94.7 | 69.2 | 51.8 | 49.6 | 42.8 | 30.8 | 56.9 | 42.5 | 21.2 | 16.6 | 57.9% | 66.9% | 68.9% | 0.3% | 0.3% |
| TEL | TOELF | C-1-7 | 72,940 | 210.2 | 58.3 | 48.9 | 39.7 | 16.2 | 13.7 | 11.4 | 33.2 | 27.3 | 10.6 | 9.1 | 29.3% | 30.3% | 31.4% | 1.0% | 1.3% |
| Hanmi Semi | HNSIF | C-1-7 | 251,000 | 13.4 | 111.2 | 79.6 | 41.6 | 34.7 | 26.1 | 17.3 | 61.7 | 30.8 | 30.2 | 16.9 | 34.8% | 37.3% | 49.8% | 0.4% | 0.8% |
| BESI | BESVF | B-1-7 | 281.60 | 26.0 | 167.1 | 65.3 | 39.1 | 53.4 | 39.1 | 28.6 | 49.7 | 31.2 | 22.5 | 16.0 | 29.2% | 69.4% | 84.6% | 1.4% | 2.5% |
| ASMPT | ASMVF | C-1-7 | 194.20 | 10.4 | 76.6 | 48.8 | 29.4 | 4.7 | 4.5 | 4.1 | 27.8 | 18.2 | 4.2 | 2.9 | 6.5% | 9.4% | 14.5% | 1.0% | 1.4% |

**Source:** Company reports, BofA Global Research estimates; Mcap = market capitalization, ROE = return on equity, Div = dividend. RSTR = Restricted. Price as of latest close.

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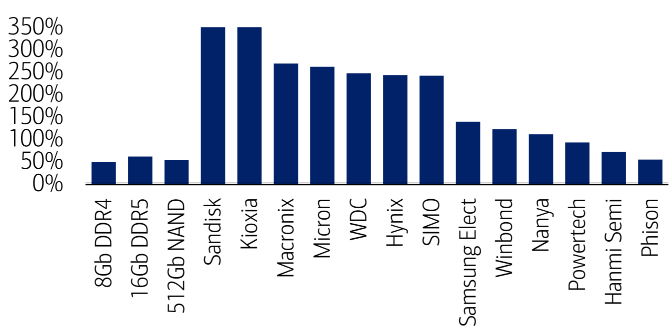
**Exhibit 40: Memory stocks corrected sharply this week after a strong 1H26 rally; however, prices still remain elevated year-to-date, led by NAND and HDD with sustained strength in DRAM, underpinned by robust AI-driven demand**

Memory companies – 2026 YTD stock performance comparison




SanDisk/Kioxia shows above 800% YTD return

**Source:** Bloomberg, DRAMeXchange

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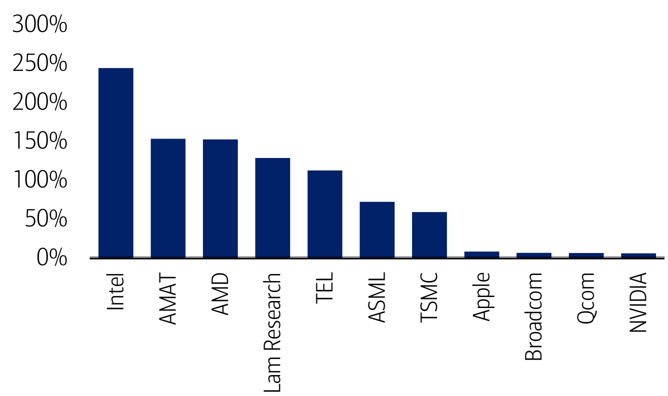
**Exhibit 41: CPU names such as Intel and AMD clearly outperforming other US big tech names such as NVIDIA/Apple/QCOM**

Global major tech companies – 2026 YTD stock performance comparison




**Source:** Bloomberg

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| Investment rating | Total return expectation (within 12-month period of date of initial rating) | Ratings dispersion guidelines for coverage cluster[R1] |
|---|---|---|
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| Neutral | ≥ 0% | ≤ 30% |
| Underperform | N/A | ≥ 20% |

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